

App Dev 1 Project Report

Student Details

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About Me:

I am an undergraduate student pursuing a B.S in Data Science with an interest in software development. I have experience with Python, JavaScript, Flask, and SQL, and I enjoy building practical applications. As part of my coursework, I developed a placement portal that streamlines recruitment workflows for students, companies, and administrators. I am interested in software development and continuously work on improving my problem-solving and programming skills.

Project Details

Project Title: Placement Portal Application.

Problem Statement:

The project aims to develop a Placement Portal Application to centralize and streamline campus recruitment activities. It enables Admin, Company, and Student roles to efficiently manage placement drives, applications, and approvals, replacing inefficient manual processes.

Approach:

The project was developed by identifying key roles, Admin, Company, and Student, and designing a role-based system. A SQLite database was used to manage all entities and relationships. Flask handled the backend logic using Blueprints for modular structure, and Flask-WTF was used for form handling and validation. The frontend was built using Jinja2 and Bootstrap.

Core features like authentication, approvals, and application tracking were implemented to ensure efficient and reliable functionality.

AI/LLM Declaration

I used ChatGPT and Claude to assist with generating SQLAlchemy model definitions and improving UI responsiveness. The extent of AI/LLM usage was approximately 10–15%. All core implementation, debugging, and system integration were completed independently.

Technologies and Frameworks Used

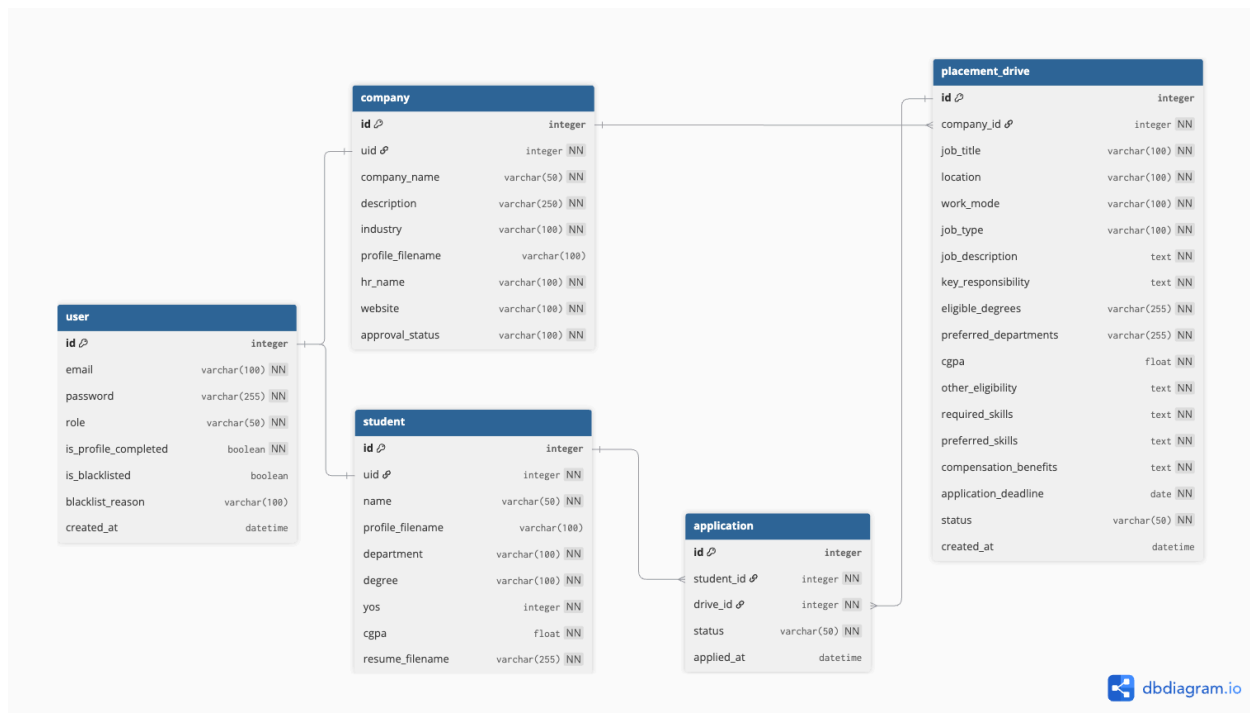
Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Flask-Login	User authentication and session management
WTForms	Frontend form validation
SQLite	Lightweight local database for storing user data

Database Schema / ER Diagram

Tables:

- **User:** Stores user details (id, email, password, role, profile status, blacklist info, etc.)

- **Company:** Stores company information (name, description, industry, HR details, website, approval status, etc.)
- **Student:** Stores student details (name, department, degree, CGPA, resume, profile info, etc.)
- **Placement_Drive:** Stores placement details (job title, location, eligibility, skills, deadline, status, etc.)
- **Application:** Stores student applications to drives (student, drive, status, applied date, etc.)



Architecture and Features

Architecture Overview:

- **routes:** Flask Blueprints are used to organize the application into modular components based on user roles (Admin, Student, Company), improving readability and scalability. Blueprints are registered in `__init__.py`.

- **app.py:** Initializes the Flask app, SQLite database, and predefined admin user. It also includes utility functions for updating drive status and handling applications of blacklisted users.
- **auth_decorator.py:** Implements role-based access control using custom decorators to restrict route access based on user roles.
- **config.py:** Manages configuration variables like SECRET_KEY and database URI using environment variables via dotenv and os.
- **forms.py:** Handles form creation and validation using Flask-WTF.
- **models.py:** Defines all database models.
- **utils.py:** Contains helper functions used across the application.

Features:

- User registration and login.
- Secure password hashing for enhanced security.
- Intuitive and user-friendly dashboard.
- Robust form validation using Flask-WTF.
- Route protection with access control
- Responsive UI.

Video Presentation

Drive link:  24f2005600 Project Demo.mov